

Create a simple Sales and Billing Management System for a college-level project.

The system should include:

1. Customer details – Customer ID, Name, Phone Number, Address
2. Product details – Product ID, Product Name, Price
3. Quantity – Number of products purchased
4. Sale – Record the products sold to the customer
5. Bill calculation – Calculate Total Amount = Price × Quantity

Prepare the following in a simple and easy-to-understand format:

1. Requirements

- List the functional requirements.
- List the data required.
- Mention the input, processing, and output.

2. Algorithm

Write a step-by-step algorithm for:

- Entering customer details
- Entering product details
- Entering quantity
- Calculating the bill
- Displaying the final bill

3. Flowchart

Create a clear flowchart using standard symbols:

- Start/End
- Input/Output
- Process
- Decision, if required

Show the complete flow from entering customer and product details to calculating and displaying the bill.

4. ER Diagram

Create a simple ER diagram with these entities:

- CUSTOMER
- PRODUCT
- SALE
- BILL

Show suitable attributes, primary keys, and relationships between the entities.

Keep everything simple, clear, and suitable for BCA students. Use easy language and avoid unnecessary complexity.